First name: Last name: Group:	ensia The National School of Artificial Intelligence المدرسة الوطنية العليا للذكاء الاصطناعي

Web Development Final Exam 2024/2025

Exercise 1: PHP/MySQLi - (11 pts)

1) Using PHP/MySQLi, connect to a database named **ENSIA_DB**, hosted on a server named **ENSIA_SRV**, with a username **ENSIA_USR** and password " **123456** ". You should echo **"Successfully connected"** if everything is OK or **"Unable to connect"** in case of a problem.

File name: connect.php

```
if (.....                                                    .....) {

..                                                                  .....

.....                                                                .....

} else {                                                            .....}
```

2) **ENSIA_DB** contains the following tables:

USER_TABLE (USER_ID, NAME, EMAIL, PASSWORD),

PRODUCT_TABLE (P_ID, P_NAME, P_PRICE, P_DESCRIPTION).

Given the following form:

```
<form action="login.php" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br><br>
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br><br>
    <input type="submit" value="Login">
</form>
```

Complete the PHP code for user authentication and redirect to the **products.php** page.

Note: name and email should be saved using sessions.

File name: .....

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```
if ($_SERVER["REQUEST_METHOD"] == "POST") {

$sql = "SELECT * FROM USER_TABLE WHERE EMAIL=? AND PASSWORD=?";

$stmt->fetch();
    if ($stmt->num_rows > 0) {

    } else {  echo "Login failed!";  }
```

3) Using the USER_TABLE and the following form, write a PHP code to register users into that table, then go back to index.php

```
<form action="register.php" method="post">
    <label for="new_username">New Username:</label>
    <input type="text" id="new_username" name="new_username" required><br>
    <label for="new_password">New Password:</label>
    <input type="password" id="new_password" name="new_password"
required><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required><br><br>
    <input type="submit" value="Register">
</form>
```


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File name:.

```
..
...
if ($_SERVER["REQUEST_METHOD"] == "POST") {
...
...
...
...
...
    $sql = "INSERT INTO USER_TABLE (NAME, EMAIL, PASSWORD) VALUES (?, ?, ?)";
...
...
...
...
...
}
```

4) The **products.php** page should display the list of products from the **PRODUCT_TABLE** in a **<table>**. The user information (name and email) should be displayed with **<h1>** tags above the table. In addition, each row must have a **delete** button.

```
...
...
...
```


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```
if (isset(                                                    )) {

}

    $sql = "SELECT * FROM PRODUCT_TABLE";

    $result =

    echo "<table><tr><th>Product ID</th><th>Product Name</th> <th>Price</th>
<th>Description</th><th>Action</th></tr>";

    while ($row = $result->fetch_assoc()) { //

echo "<td><button
onclick='deleteProduct("                    ")'>Delete</button></td>";

}

    echo "</table>";
```

5) Users can change text and background colors using the given form on the **settings.php** page. Complete the PHP code that saves those settings in cookies and use them to change the **<body>** background and text colors.

```
<form action="settings.php" method="post">
        <label for="BG_Color">Background color</label>
        <select name="BG_Color">
            <option value="green">Green</option>
            <option value="yellow">Yellow</option>
        </select>
        <label for="Text_Color">Text color</label>
        <select name="Text_Color">
```


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```
            <option value="green">Green</option>
            <option value="yellow">Yellow</option>
        </select>
        <input type="submit" value="Save">
    </form>
// save settings in cookies
if (isset(..                                        ..))  {
..                                                          }
if (isset(..                                                ..))  {
..                                                          }
// change <body> background and text colors
<body style="background-color: <?php if
(isset(..                                                   ..))  {
..                                                          }  ?>;
  color: <?php if
(isset(..                                                   ..))  {
..                                                          }  ?>;">
```

Exercise 2: PHP/AJAX - (6 pts)

Consider the following JS and PHP source code:

1- The first set contains functions to load products from a php file (products.php) and the second to delete a product using the product id and send it to the php server,

2- The second set contains the php scripts that select products and remove the product after receiving the product id.

Assumptions:

1- There are no errors in the source code

2- We removed the <html> and <head> tags.

```
<body>
      <h1>Product List</h1>
      <div id="product-list"></div>
      <script>
      function loadProducts() {
```


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```
        const xhr = new (1)                                    ...................................;
        xhr.open('GET', (2) ...                                ...................................., true);
        xhr.onload = function () {
            if (xhr.status === (3)..                           ...................................) {
                document.getElementById('product-list').innerHTML =
xhr.responseText;
            } else {
                alert('Failed to load products.');
            }
        };
        xhr.onerror = function () {
            alert('Failed to load products.');
        };
        (4).                                                   ...................................;
    }
    function (5)..... ..... ..... .....(productId) {
        const xhr = new (6)....                                ...................................;
        xhr.open((7)..                                         ...................................., 'products.php', true);
        xhr.setRequestHeader('Content-Type',
'application/x-www-form-urlencoded');
        xhr.(8)...                                             ..................................... = function () {
            if (xhr.status === 200) {
                alert(xhr.responseText);
                (9).....                                       ..................();
            } else {
                alert('Failed to delete the product.');
            }
        };
        xhr.onerror = function () {
            alert('Failed to delete the product.');
        };
xhr.send(`action=(10)..                                        ..&(11)..                    .....=${productId}`);
    }
    document.(12)..                                            .....('DOMContentLoaded', () => {
        loadProducts();
        document.addEventListener('click', event => {
            if (event.target.classList.contains('delete-btn')) {
                const productId = event.target.getAttribute('data-id');
                if(confirm('Are you sure you want to delete this product?'))
                {
                    (13)....                                   .....((14)....                    .....);
                }
            }
        });
    });
    </script>
</body>
```

And the PHP source code

```
<?php
$host = 'localhost';
$dbname = 'your_database';
$user = 'your_username';
$pass = 'your_password';
try {
    $pdo = new PDO("(15)                                       ", $user, $pass);
    $pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
} catch (PDOException $e) {
```


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```
die("Database connection failed: " . $e->(16)..                    .....);
}

if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) &&
$_POST['action'] === 'delete') {
      $productId = intval($_POST[(17)....          .....]);

      $stmt = $pdo->prepare("DELETE FROM products WHERE id = ?");
      if ($stmt->(18)..          .....([(19)..          .....])) {
      echo "Product deleted successfully.";
      } else {
      echo "Failed to delete product.";
      }
      exit;
}

if ($_SERVER['REQUEST_METHOD'] === (20).....          .....) {
      $stmt = $pdo->(21)...          .....("SELECT id, name FROM products");

      $html = '<ul>';
      while ($row = $stmt->fetch((22)....          .....)) {
      $html .= "<li>{$row['name']}
                        <button class=(23).          .....
data-id='{$row[(24)          .....]}'>Delete</button></li>";
      }
      $html .= '</ul>';
      echo $html;
      exit;
}
?>
```

Complete the missed parts in the source codes to make them functional.

Exercise 3: JS - (3 pts)

Complete the Javascript code to validate the following form:

```
<form>
        <label for="Email">Email</label>
        <input type="email" id="Email" name="Email" required> <br>
        <label for="Phone">Phone</label>
        <input type="tel" id="Phone" name="Phone" required> <br>
        <input type="submit" value="Submit" id="submit">
```


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```
    </form>
    <div id="error_msg"></div>
```

JS code

```
submit = ..                                                    ..........

Email =                                                        ..........

Phone = ..                                                     ..........

error_msg = ..                                                 ..........

submit.....(".....                    .....", (event) => {

....                                                    .....;

if (
!/^([a-zA-Z0-9_\-\.].....  .....).....  .....([a-zA-Z0-9_\-\.].....  .....)\.([a-zA-Z]
{.....                    .....})$/.test(.....                    .....)) {

    error_msg.innerHTML += "Email is not valid \n";   }

  if (!/^([        .....]{.....  .....}[...                .....]{.....  .....}[.....]

  {.....})$/.test(.....                    .....)) {

 error_msg.innerHTML += "Phone is not valid \n";   } });
```

Don't forget to review your sheet before leaving

Don't forget your first name, last name and group

Good luck

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